Answers

Starter quiz

Converting terminating decimals to fractions

1 A decimal with an infinite repeating pattern of digits is called a ___________ decimal.
Tick **1** correct answer

- [ ] fractional
- [ ] proper
- [x] recurring ✓
- [ ] terminating

2 Match each decimal given using dot notation to its equivalent decimal. Write the correct letter in each box

a	$0.\dot{7}\dot{8}$
b	$0.78\dot{9}$
c	$0.\dot{7}8\dot{9}$
d	$0.\dot{7}$
e	$0.7\dot{8}\dot{9}$

c	$0.789789789...$
d	$0.77777777...$
e	$0.789898989...$
b	$0.789999999...$
a	$0.78787878...$

3 Select all the fractions which are equivalent to a recurring decimal. Tick **3** correct answers

- [x] $\frac{5}{12}$
- [ ] $\frac{6}{24}$
- [x] $\frac{2}{3}$
- [ ] $\frac{9}{15}$
- [x] $\frac{2}{12}$

4 Use short division to write $\frac{11}{15}$ as a decimal. Tick **1** correct answer

- [ ] 0.73
- [ ] 0.73333333333
- [ ] $0.\dot{7}\dot{3}$
- [x] $0.7\dot{3}$

5
Lucas uses his calculator to write $\frac{7}{18}$ as a decimal. His calculator display shows the number 0.3888888889. Lucas says, "This shows my fraction terminates." Is Lucas correct? Explain why. Tick **1** correct answer

- ☐ Yes; Lucas' number has 10 digits after the decimal point so it terminates.
- ☐ Yes; the last digit is 9 not 8 so it doesn't repeat. So it must terminate.
- ☐ No; all decimals recur eventually.
- ☑ ✓ No; the calculator only has a 12 digit display, the last 8 is rounded up to 9.

6
Aisha writes the recurring decimal $0.61\dot{6}$ as the fraction $\frac{\square}{60}$. What number should she write in the square? 37 Fill in the blank